Choosing the Medium of Neural Reasoning

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When does expressing intermediate work in language help a neural system reason, and when do structured or continuous representations support better inference? I propose to investigate this question within individual models, focusing on computations that must retain alternatives and revise conclusions when new evidence arrives. By combining controlled tasks, interventions on intermediate states, and comparisons across resource budgets, I would identify which distinctions each medium preserves and actually uses. The aim is to explain when a medium improves reasoning and when an apparent advantage comes from extra computation, memory, or easier training.

Motivation and research gap

A partial solution may need to preserve several possible assignments and the evidence supporting them. A concise verbal explanation may omit such distinctions, but language can also provide a stable, compositional workspace. The question is which properties of a task make a medium useful under finite resources. A textual final answer does not imply that the preceding computation was linguistic: ordinary transformers retain hidden states and access to their inputs.

Coconut feeds continuous hidden states back into a language model [1], and recurrent-depth models allocate computation without generating more tokens [2]. Recent studies already examine causal influence, alternative answer modes, and failures of sequential latent computation [3,4]. Qiu et al. further distinguish content-dependent latent reasoning from chains that merely provide computational slots [5]. Meanwhile, Markovian language models demonstrate the importance of specifying the actual information boundary [6]. I would build on this work by testing whether a medium preserves information needed for later revision, and whether task structure predicts its advantage after accounting for computation and memory.

Research questions

Which intermediate distinctions remain usable in verbal, structured symbolic, and continuous states as branching, depth, and uncertainty increase? When do differences in this retained information explain success on later inference, rather than differences in available compute or input access? Here, a medium is the representation carried between computational updates. Useful retention means that the model can use a distinction to make a correct subsequent decision; decodability by a separate probe is insufficient.

Proposed approach

Start with reasoning that requires revision. I would generate relational constraint problems with known sets of consistent assignments. Early evidence would leave several possibilities open; later evidence would eliminate some and require a revised answer. Branching, dependency depth, and evidence order could be varied independently. Matched early states would support the same current answer but require different answers after a shared new clue. This provides a direct test of information that matters beyond the answer already available. Exact symbolic solvers would supply reference states and correctness checks.

Compare media within a shared model family. Using an open backbone, I would train variants with natural-language scratchpads, structured symbolic states such as assignment tables, and Coconut-style continuous steps. Each would receive the same task examples and comparable tuning effort, with supervision differences explicitly recorded. Text baselines would be trained to retain uncertainty and alternatives, rather than restricted to a single guessed path. A filler-token control would test benefits from additional computational steps without informative surface content [7]. A second backbone or recurrent architecture would be a later replication, not a prerequisite.

Separate native reasoning from an explicit information boundary. First, I would compare the variants with their normal prompt and cache access. In a complementary interruption-and-continuation experiment, the same model would resume from a verbal, symbolic, or latent encoding of a matched partial computation. Its earlier prompt and cache would be removed; only the retained state, task instructions, and new evidence would remain. Learned encoders and decoders would be trained for continuation and compared with equal-capacity latent round-trip controls. This isolates what survives conversion while acknowledging that an artificial reset is different from ordinary token generation. A loss observed only after reset would support an interface limitation, not a claim that all linguistic reasoning is constrained in that way.

Test why a medium succeeds. Probes would locate candidate encodings of alternatives and their dependencies. Content substitutions between matched problems, selective state edits, and deletion controls would test whether the continuation actually uses them. A proposed explanation must predict the direction of answer changes on held-out structures. I would also test whether supplying an omitted distinction in a structured text state rescues performance. This separates failure to preserve information from inability to use it, and avoids interpreting every latent-state perturbation as evidence of meaningful reasoning.

Compare practical trade-offs. I would report accuracy and revision success against inference FLOPs, latency, and retained-state storage, including encoding and decoding costs. Equal token counts would not be treated as equal information or computation. Training cost and finite-precision latent storage would be reported separately. Budget sweeps would identify regions where one medium is useful; larger-budget controls would test whether an apparent limitation is a resource tradeoff. The goal is not to prove that finite latent information is intrinsically inexpressible through symbols.

Feasibility and expected contribution

The first three months would establish one constraint family and strong verbal and symbolic baselines. Months four to eight would add continuous computation and validate the interruption experiment. Months nine to twelve would test causal explanations and generalization across task structures. Small models and bounded tasks keep the initial programme tractable; if latent training is unstable, the state-conversion study remains a focused investigation of representational loss.

The expected contribution is an account of which computational demands favor each medium, supported by a benchmark and causal tests. It could guide the design of reasoning workspaces and later hybrid systems. Communication would follow this result: work such as Swarm Telepathy [8] offers a route to test whether another model can use the retained state. Human-facing extensions could expose it through diagrams or interactive structured answers; a raw latent vector is not itself a human-readable explanation. Comparison with human verbal and nonverbal reasoning would be a separate, hypothesis-driven extension.

Selected references

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